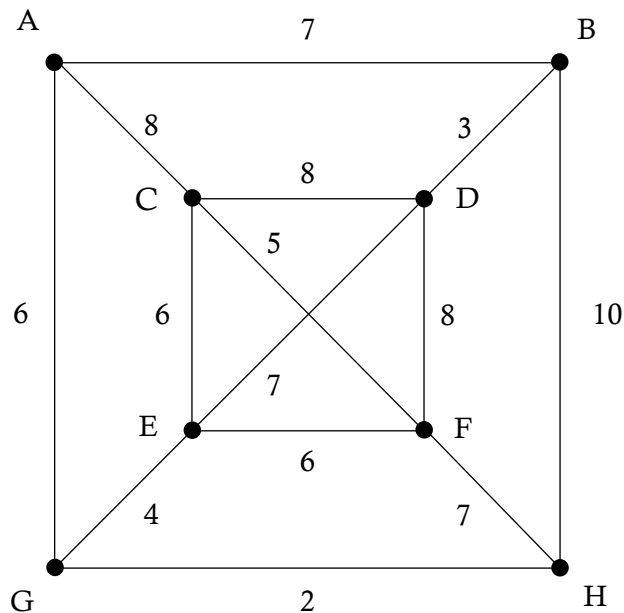


PRIM'S ALGORITHM

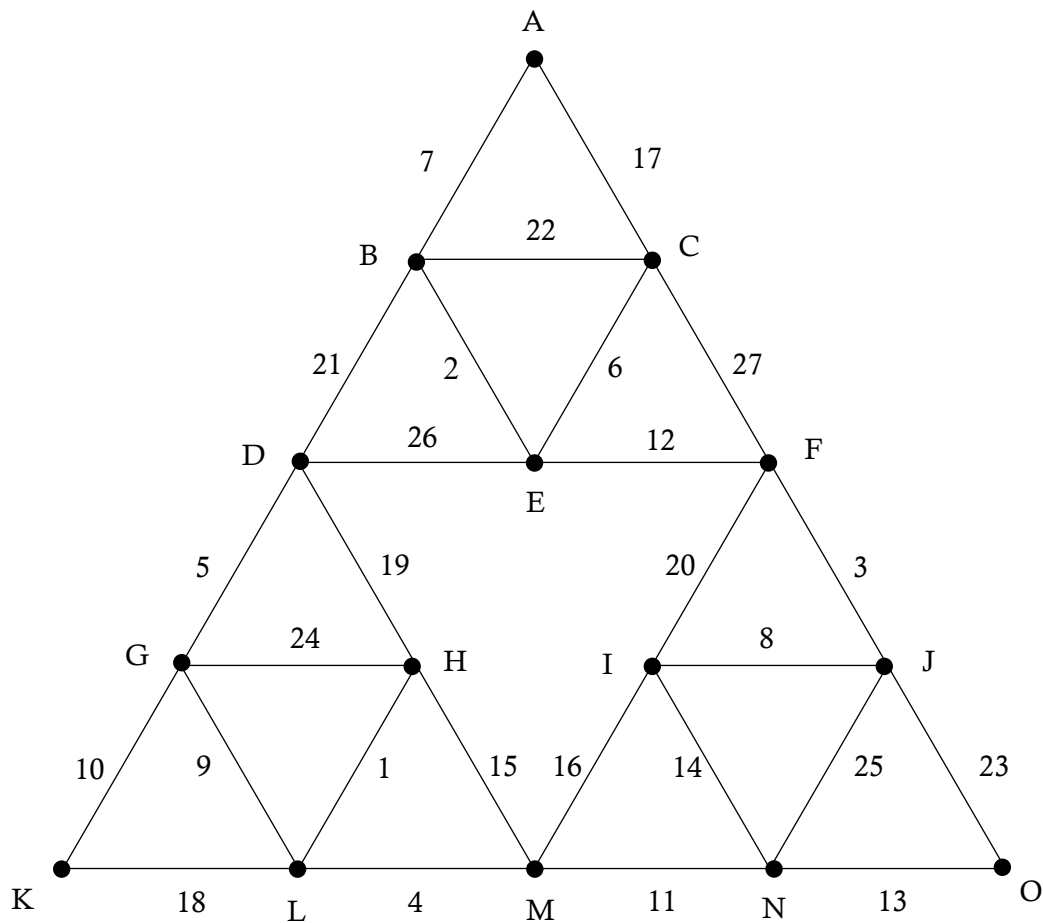
05/12/25

1. Consider the network below:



- a. Use Prim's algorithm starting at A to find a minimum spanning tree for this network. Show your working.
 - b. Show that there are four different minimum spanning trees that could have been found using Prim's algorithm starting at A, and draw these.
 - c. Show that if Prim's algorithm starts at B then there are six possible ways in which the algorithm could run, but each of these ways finds one of the minimum spanning trees you drew above.
2. Suppose that you have a complete network on five vertices, and you want to use Prim's algorithm to find the minimum spanning tree.
- a. Once you have chosen a starting vertex, how many edges will you have to consider to find the first edge of your tree?
 - b. Once you have two vertices in your tree, how many edges need to be considered to find the next edge of the tree?
 - c. How many edges will you need to consider in total while carrying out the algorithm?
3. Suppose that you have a complete network on eight vertices, and you want to use Prim's algorithm to find the minimum spanning tree.
- a. At a particular time you have a tree on five vertices. How many edges will you need to consider in order to add a sixth vertex, using Prim's algorithm?
 - b. How many edges do you need to consider in total while carrying out the algorithm?

4. Consider the network below.



- Use Prim's algorithm starting at A to find a minimum spanning tree of this network (you don't need to show your working).
- Look at the weights of the edges that are in the minimum spanning tree. What do you notice?
- Can you suggest a possible alternative algorithm to find a minimum spanning tree? Test your alternative on the networks shown below:

